

Response to the Editor and Reviewers

Construction project dependency networks compress exponentially into logical-depth levels whose mean distance to project completion is nearly independent of project size

(previously: “High-Level Structure of Dependency Networks”)

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I thank the handling editor, Dr Roy Cerqueti, and the three reviewers for their careful and constructive reading. The comments identified two genuine weaknesses — insufficient statistical support for the claimed relations, and an incomplete engagement with the neighbouring literatures — together with an uninformative title and several presentational problems. I have addressed all of them, and in doing so I have had to weaken two of the claims made in the original manuscript. Specifically:

- The number of projects analysed is 1198, not a handful. This was not stated in the original manuscript, which is why Reviewer 1 reasonably assumed the sample was small. A descriptive statistics table has been added.
- Formal model selection now supports the logarithmic form for the DUS and the sub-linear power law for the DPS, but the DUS relation explains only 27% of the variance and does not extrapolate to the public PSPLIB benchmark. I have therefore stopped calling it a scaling law and describe it as slow logarithmic growth of the central trend.
- The small-world question is now settled by the analysis Reviewer 1 asked for. Clustering is given a DAG-appropriate definition, two degree-preserving null models are built, and the result is that the short distances follow from the degree distributions of the DUS network rather than from any special arrangement of the shortcuts. The distances are short; the Watts–Strogatz criterion is simply not the informative test for these networks.
- The finer construction has been renamed from DFS to DPS (Dependency Predecessor Structure) to remove the clash with depth-first search.
- All key results are now replicated on public data (PSPLIB and two anonymised real schedules shipped with the code), and every number and figure in the paper is produced by scripts included in the public repository.
- The title has been replaced, following the editor’s comment, by a single 21-word sentence naming the system studied and the two principal findings.

The paper has been restructured accordingly: the schematic that was Figure 4 is now Figure 1, a descriptive statistics table and two new figures have been added, and the manuscript now has six figures and one table.

A correction found while preparing this revision

While rebuilding the analysis so that it could be released publicly, I recomputed every quantity directly from the raw schedules and compared the result against the summary table used for the original submission. The number of activities, the number of dependencies, the number of

DUS levels and the distances agreed exactly for all 1198 projects. The number of DPS classes did not.

The cause was a defect in the original code. The DPS label packs one bit per DUS level, and it was accumulated with `math.pow(2, j)`, which returns a float. A float64 mantissa holds 53 bits, so once a schedule has more than 53 DUS levels the low-order bits are lost, distinct predecessor patterns collapse onto the same value, and counting distinct labels under-estimates the number of DPS classes. The error is exactly zero at or below 53 levels and grows with depth: across the 1198 projects the under-count had a median factor of 1.010, rising to a median of 1.05 between 100 and 200 levels and 1.17 above 200 levels, with a worst case of 1.51.

Because the under-count grows with project size, it biased the fitted exponent downward. Every DPS result in this revision has been recomputed with exact integer arithmetic. The corrected exponent is 0.709 [0.692, 0.724] rather than 0.694 [0.679, 0.709], the prefactor is 1.42 rather than 1.56, and the DPS column of Table 1 shifts slightly. The conclusion is unchanged and, if anything, strengthened: the growth is a power law, decisively preferred over the alternatives, with an exponent well below 1. The DUS results, the distance results and Figures 3 and 4 are untouched, since the defect affected only the DPS labelling.

I report this rather than silently substituting the corrected numbers. The fix is committed to the public repository with the diagnosis in the commit message, and the released implementation now builds the label with exact integer bit shifts. It was Reviewer 1's request for public replication that prompted the rewrite which exposed the problem.

Editor

Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained.

Response. This is the axis along which the revision has changed most, and in three concrete ways. On accurate reporting: every number in the paper has been recomputed from the raw schedules, which is how the coding defect described above came to light, and each figure and statistic is now produced by a script in the public repository. On overstated conclusions: the term “scaling law” has been dropped for the relation between the number of DUS levels and the number of dependencies, which explains only 27% of the variance and fails to extrapolate to public benchmark data; the small-world claim has been replaced by what a degree-preserving null model actually supports; and where model selection formally prefers a weakly increasing relation over an exactly constant one, the manuscript says so rather than claiming constancy. On limitations: the Discussion now carries an explicit paragraph covering the single-sector sample, the non-universality of the fitted coefficients, the fact that the DUS is one antichain decomposition among many with no optimality claim, and the scope of the null models used.

The current title, “High-Level Structure of Dependency Networks”, appears to be too short and does not clearly communicate the scientific focus of the manuscript. Could you please revise the title so that it is more descriptive, scientifically informative, and accurately reflects the content of the study? Where possible, the title should be read as one concise sentence which is under 22 words long.

Response. Agreed, and the title has been changed to

“Construction project dependency networks compress exponentially into logical-depth levels whose mean distance to project completion is nearly independent of project size”

which is a single sentence of 21 words. It names the empirical system (construction project dependency networks), the principal structural result (exponential compression into levels defined by logical depth), and the second finding that the mean distance to the end of the project does not grow appreciably with project size. The qualifier “nearly” is deliberate: model selection marginally prefers a weakly increasing relation to an exactly constant one, so a title asserting strict independence would overstate the result, which the comment above asks me to avoid.

Reviewer 1

The manuscript proposes a deterministic two-step coarse-graining method for dependency DAGs. . . the current version of this paper shows some weaknesses in terms of its originality, statistical rigor, and the validation of the two arguments presented.

Response. I agree with this assessment of the original version. Each of the four points is addressed below.

1. Missing comparison with DSM, WBS/CPM and DAG condensation

From a project management perspective, the Design Structure Matrix (DSM) literature has a long history regarding the clustering and ordering of dependency matrices, and the concepts of WBS and CPM are conceptually very similar. Yet, this paper does not cite or compare these studies. From a computer science perspective, DAG condensation and hierarchical graph techniques are similarly relevant.

Response. Accepted. The Introduction now contains a paragraph that positions the work against both literatures and states explicitly what is and is not new.

On the project management side, I now cite Steward’s original DSM paper, Browning’s review and the Eppinger–Browning monograph, and I draw the distinction that matters: DSM sequencing and the layering used here both look for an ordering compatible with the dependencies, but DSM methods are typically applied to matrices that contain cycles, where the object is to find and break feedback loops, and DSM clustering is usually the outcome of an optimisation with tunable objectives and no unique answer. The construction here applies to acyclic networks, involves no optimisation and no free parameters, and is aimed at compression rather than resequencing. CPM is now cited at the point where the schedule is introduced, and the relationship is made explicit in the DUS section: the DUS index of the final activity is the number of activities on the longest dependency chain, which is the critical path when all activities have equal duration. The relation to the WBS is now a structural part of the paper rather than a passing remark — Figure 1A shows a WBS, Figure 1B the dependency network of its leaves, and the caption points out a dependency that crosses a WBS branch boundary, which is the reason the two views do not coincide.

On the computer science side I now cite and distinguish three things that might otherwise be confused with this construction. Condensation by strongly connected components (Tarjan) is the standard way to turn a digraph into a DAG; applied to a schedule, which is already acyclic, it does nothing at all. Transitive reduction (Aho, Garey and Ullman) removes redundant edges but preserves every node and therefore achieves no node-level compression. Layered graph drawing (Sugiyama; Healy and Nikolov) does assign nodes to layers by essentially Eq. (1), which the original manuscript already acknowledged, but treats the layering as a step towards a picture rather than as an object of study.

I want to be straightforward about the resulting claim of novelty, which is narrower than the original manuscript implied. The DUS index is the standard longest-path layering and I do not

claim it as new. What I claim is: its use as a coarse-graining with measured compression, the empirical relations of Figures 3 and 5 with the model selection now supporting them, the DPS refinement, and the observation that the resulting networks are shallow. This is now stated in those terms at the end of the DUS section.

2. Statistical support for the scaling relations

(i) The logarithmic fit in Figure 2 appears to be a single-parameter fit; the sample size, a goodness-of-fit measure, and a confidence interval on p should be reported. . . a model-selection argument (e.g., AIC/BIC or a likelihood-based comparison) is needed to justify the logarithmic form.

Response. Accepted in full, and this changed the results. The analysis has been redone as follows. Four functional forms (one- and two-parameter logarithms, a power law, a linear relation) were each fitted by maximum likelihood under two error structures (additive and multiplicative), giving eight fits. The multiplicative log-likelihoods include the Jacobian term $-\sum_i \ln y_i$ so that all eight are likelihoods of the same observed quantity and their AIC values are directly comparable across both functional form and error structure. This is described in a new Statistical methods subsection.

The outcome supports the logarithmic form, but with a caveat the original manuscript did not report:

- The preferred model is the two-parameter logarithm with multiplicative errors, $m = 14.3 \ln x - 33.9$. The reviewer was right that the published fit was a one-parameter logarithm; that form is now rejected by $\Delta\text{AIC} = 45$.
- It is preferred over the power law by $\Delta\text{AIC} = 22$ and over the linear relation by $\Delta\text{AIC} = 147$. The best-fitting power law has exponent 0.17, which as the reviewer anticipated is small enough that the two forms are hard to separate visually; the likelihood comparison is what separates them.
- $n = 1198$. The 95% bootstrap confidence interval on the logarithmic coefficient is $[13.4, 15.2]$. Residuals show no detectable heteroscedasticity (Breusch–Pagan $p = 0.07$).
- Goodness of fit is modest: $R^2 = 0.27$ on the logarithm. This is now stated in the text, and Figure 3 shows binned medians with interquartile ranges so that the reader can see the scatter directly.

Given $R^2 = 0.27$, and given the extrapolation failure described under point 4 below, I have removed the term “scaling law” for this relation and describe it instead as slow logarithmic growth of the central trend. The compression claim survives and is now quantified concretely: a tenfold increase in dependencies adds about 33 DUS levels regardless of starting size.

(ii) The power-law fit in Figure 5 should specify the fitting method; a naive log-log OLS is biased, so a Clauset–Shalizi–Newman-style procedure (Clauset et al. 2009), or at least residual diagnostics and a confidence interval on the exponent, would be appropriate.

Response. The fitting method is now specified in full, and residual diagnostics and a confidence interval are reported. The preferred model is a power law under multiplicative errors, $1.42 x^{0.71}$, ahead of the linear relation by $\Delta\text{AIC} = 269$ and the two-parameter logarithm by $\Delta\text{AIC} = 1310$. The exponent is 0.709 with a 95% bootstrap confidence interval of $[0.692, 0.724]$, comfortably excluding 1, and $R^2 = 0.81$ on the logarithm. The residuals of this fit are heteroscedastic, which is now stated explicitly; this is precisely why the interval quoted is a nonparametric case-resampling bootstrap rather than a nominal standard error.

On the Clauset–Shalizi–Newman procedure I would respectfully disagree, and have added a short paragraph to Methods explaining why. That procedure fits and tests a heavy-tailed *probability distribution* of a single random variable, using maximum likelihood for the tail exponent above an estimated $x_{\min}$ together with a Kolmogorov–Smirnov goodness-of-fit test. The relation in Figure 5 is not a distribution: it is a regression of one measured quantity (number of DPS classes) on another (number of dependencies) across projects. The estimator, the $x_{\min}$ selection step and the KS test all presuppose i.i.d. draws from a distribution and have no direct analogue here. The reviewer’s underlying concern — that naive log-log OLS is not a neutral choice — is entirely valid, and I have addressed it in the way that is appropriate for a regression: by making the error model an explicit part of the model comparison rather than an unstated assumption, and by using a bootstrap that does not assume homoscedasticity. I would of course reconsider if the reviewer thinks a distributional formulation is called for.

(iii) *The claim that all three companies follow the same relation should be supported by a test of parameter commonality rather than visual impression. When the sample size is small (only a few projects per company), the use of strong terms such as “scaling law” should be used with appropriate moderation.*

Response. Accepted, and this is now tested formally. I should first correct a factual premise, and the fault for it is mine: the sample is not small. It comprises 1198 projects — 82, 249 and 867 from the three companies. The original manuscript said only that “each company contributed several projects”, which was uninformative and invited exactly this reading. The new Table 1 gives the full breakdown.

Commonality is tested with nested likelihood ratio tests comparing three models: fully pooled parameters; a shared logarithmic coefficient or power-law exponent with company-specific offsets or prefactors; and fully company-specific parameters. This separates the question of whether the companies share the *form* of the relation from whether they share its nuisance parameters, and the two answers turn out to be different:

- DUS: a shared logarithmic coefficient is not rejected (LR = 2.4, df = 2, $p = 0.30$; shared value 15.1), and AIC prefers the shared-coefficient model. The offsets do differ ($p = 5 \times 10^{-9}$).
- DPS: a shared exponent is not rejected (LR = 1.1, df = 2, $p = 0.59$; shared value 0.72), while the prefactors differ ($p = 4 \times 10^{-6}$).

So the original claim that “the same relation describes all three companies” is supported in the sense that matters — the rate of growth is common — but was too strong as stated, because the intercepts and prefactors are significantly company-specific. The revised text says exactly this and interprets the difference as company-specific scheduling practice shifting the absolute level without changing the rate.

On terminology I have followed the reviewer’s advice throughout. The phrase “scaling law” no longer appears anywhere in the manuscript as a claim. The DUS relation is described as slow logarithmic growth of the central trend, for the reasons above. The DPS relation is described as a sub-linear power law, which is a statement about functional form rather than a claim of universality, and it is the better supported of the two: a power law is preferred by a very large margin, the exponent is tightly determined, $R^2 = 0.81$, and the relation extrapolates successfully to independent public data.

3. The small-world claim

The evidence supporting the small-world claim is weak... this argument simply shows that the average distance is less than the expected value for a pure chain. Since any shortcut reduces the distance compared to a chain, this is almost a trivial statement. The Watts–Strogatz definition requires not only short path lengths but also high clustering compared to degree-preserving random graphs... the authors would need to clearly specify how clustering is defined in a DAG.

Response. Accepted on both counts, and the analysis the reviewer asks for has now been carried out. Falling below the chain expectation is indeed nearly tautological, and the original manuscript did not make the clustering comparison. The reviewer is also right that clustering needs defining first: the standard coefficient counts closed triangles, and a *cyclic* triangle cannot exist in a DAG.

That last point has a clean resolution, which the revision adopts. A cyclic triangle is impossible in a DAG, but a *transitive* one is not: $i \rightarrow j$, $j \rightarrow k$ and the shortcut $i \rightarrow k$ are mutually consistent with acyclicity. We therefore use the transitive clustering coefficient C , the fraction of directed two-paths $i \rightarrow j \rightarrow k$ closed by an edge $i \rightarrow k$. It is the admissible member of the directed-triangle family, it is precisely the quantity that a shortcut creates, and it is zero for a pure chain. This is now stated in the Results and defined in a new Methods subsection.

Two degree-preserving null models were then built, both by double-edge swaps. The first holds the DUS spine fixed, since Eq. (1) forces it and randomisation should not be free to destroy it, and rewires the skip dependencies while preserving every level’s skip in- and out-degree. The second preserves the in- and out-degree of every *activity*, keeps the network acyclic by requiring every edge to point forward in a fixed topological order, and re-derives the DUS decomposition from scratch. Mixing was checked in both cases.

The result, over 198 projects, is that the observed clustering and the observed distances are almost exactly those of the first null: $C/C_{\text{null}} = 1.04$ (IQR [1.01, 1.10]) and $L/L_{\text{null}} = 1.03$ (IQR [1.00, 1.08]). This is a new Figure 7 and a new Results subsection.

I want to be careful about what this does and does not show, because it is easy to misread it as a negative result. It does *not* cast doubt on the networks having small average distances — that is a measured property of the data, and no null model can remove it: the mean distance to the end is 2.5 levels where a chain of the same depth would give up to 285. What it shows is *why* the distances are small. Given how many long-range dependencies each level has, placing them anywhere at all reproduces the observed distances, so the shallowness follows from the degree distributions of the DUS network rather than from any special arrangement of the shortcuts. That is a mechanistic explanation, and a positive one. It also connects to the generative side of the problem: the duplication-split model of activity networks (Vazquez et al., Sci. Rep. 13, 509, 2023) produces degree distributions of exactly this kind, and the present result establishes that such distributions are sufficient for the effect.

What the analysis does bear on is the Watts–Strogatz *criterion* specifically, which asks for clustering substantially in excess of a degree-matched null. That excess is absent, so the criterion is not the informative test for these networks, and the manuscript states the property directly rather than invoking the label. This is a statement about which diagnostic is appropriate, not about whether the distances are short.

The second null adds a complementary finding. Preserving only the activity degree sequence and re-deriving the decomposition gives DUS networks roughly half as deep as the real ones (median $m/m_{\text{null}} \approx 2$, a conservative figure, since this randomisation mixes slowly and the gap widens as it proceeds). So the degree distributions determine how short the paths through the

DUS network are, but not how deep it is; real schedules retain genuine sequential structure that the degree sequence alone does not capture. The Discussion now poses the generative origin of that depth as the open question, in place of the definition of clustering, which is settled above.

The distance evidence itself has also been strengthened beyond the chain comparison, because the interesting fact turned out not to be the gap but the flatness:

- The mean distance to the final DUS level has median 2.5 with interquartile range [1.9, 3.6], essentially independent of the number of levels m , which ranges from 14 to 571. Binned medians are 3.4, 2.3 and 2.7 for small, medium and large projects.
- The rank correlation between distance and m is $\rho = 0.10$ ($p = 8 \times 10^{-4}$, $n = 1198$): nominally significant at this sample size but negligible in magnitude. For completeness the manuscript also reports that the same model selection procedure formally prefers a weakly increasing linear relation over a strictly constant one ($\Delta\text{AIC} = 20$, slope 0.0041 per level, 95% CI [0.0018, 0.0062]). The implied growth is negligible: across the whole range the fitted distance rises from 2.5 to 4.7, a factor of 1.9, while m grows by a factor of 41. I preferred to report this rather than claim exact constancy.
- Figure 4 is now on logarithmic axes, which makes the contrast between the flat data and the rising chain expectation directly visible; the original linear axes obscured it.

The word “small-world” now appears only where the shortcut mechanism is attributed to Watts and Strogatz, and where the Watts–Strogatz criterion is discussed and set aside for the reason given above, never as an unexamined label.

4. Proprietary data, descriptive statistics and public replication

It would be advisable for the paper to include at least descriptive statistics—namely, the number of projects per firm, the range of activity and dependency counts, and basic summary information on the distribution. Furthermore, reproducing key scalability results using publicly available scheduling datasets (e.g., PSPLIB) would significantly enhance the replicability of the research arguments.

Response. Both accepted and done.

Table 1 now reports, per company and overall, the number of projects and the median and full range of the number of activities, dependencies, DUS levels and DPS classes. The dataset spans 37 to 104,981 activities and 36 to 150,692 dependencies. The Methods section also now records the one schedule that was excluded from the original analysis and why (its first DUS level connects to every other level, so its DUS network is a star rather than a network with a spine); this exclusion was applied but not reported in the original manuscript.

For replication I have analysed all 2040 single-mode PSPLIB instances (j30, j60, j90, j120), plus the two anonymised real schedules that are distributed with the code repository and which extend the publicly reproducible range to 1.3×10^5 dependencies. This is a new figure (Figure 6) and a new Results subsection. The result is mixed, and I report it as such:

- PSPLIB instances have 32–122 activities and 48–257 dependencies, come in only four activity counts, and span barely half a decade in size. Model selection within PSPLIB alone favours a power law for the DUS and a linear relation for the DPS, which are not the forms selected on the construction data; over so narrow a range I do not treat that as evidence about functional form, and the manuscript reports it as such rather than presenting a within-PSPLIB fit as a confirmation.

- What PSPLIB can test is transfer. Its dependency range overlaps only the extreme lower tail of ours — just 2.4% of the construction projects are that small — so it probes the relations where they were barely constrained. The DPS power law, fitted on construction data and evaluated two orders of magnitude below the median project on which it was estimated, predicts the PSPLIB values to a median observed/predicted ratio of 0.77. The power law fitted within PSPLIB alone gives exponent 0.86, also sub-linear. This is a genuine out-of-sample success.
- The DUS logarithm does not extrapolate: it over-predicts PSPLIB by more than a factor of two (median ratio 0.42). The paper now states that the logarithmic relation describes large construction schedules rather than being universal, which is the second reason for dropping the term “scaling law” for it.
- The qualitative shallowness result does replicate: median distance to the final level 2.1, below the chain expectation in 100% of instances.

Finally, the code repository now contains the scripts that perform every statistical test, download and process PSPLIB, and generate every figure and the table in the paper, so that all public-data results can be reproduced end to end.

Reviewer 2

Both methods reveal a small-world structure in the underlying dependency network... I would like to highlight two main limitations.

Response. I thank the reviewer for an accurate summary. Note that the small-world claim has been withdrawn following Reviewer 1’s point 3; see that response.

1. Structure and placement of illustrative material

The structure of the paper is unconventional, and key illustrative material is placed too late. For instance, the illustration of the two metrics appears in Figure 4, whereas it would be far more effective as Figure 1, orienting the reader from the outset.

Response. Accepted. The schematic is now Figure 1 and appears in the Introduction, before either construction is defined. It has also been substantially redrawn and expanded to three panels so that it can carry more of the exposition: panel A is a small Work Breakdown Structure, panel B the dependency network of its leaf activities with the DUS index shown as vertical bands and the DPS class shown by colour, and panel C the two coarse-grained networks. The example was chosen so that the DPS genuinely refines the DUS — two activities share a DUS level and a DPS class while a third shares their level but forms its own class — which the original schematic did not illustrate. Panels A and B also make visible the point that dependencies cross WBS branches.

The figure order is now Figure 1 schematic, Figure 2 real DUS profiles, Figure 3 DUS scaling, Figure 4 distances, Figure 5 DPS scaling, Figure 6 public replication, plus Table 1.

I would introduce DAGs more extensively, and move the first paragraph of the Results section into the Introduction.

Response. Both done. The first paragraph of Results has been moved into the Introduction and expanded into a fuller treatment of DAGs: the definition, the acyclicity requirement and why it is forced, topological order, the notion of incomparability that the DUS relies on, and

a note that DAGs also represent causal models, ancestries and feedforward networks, which is what makes the constructions transferable. The Results section now opens directly with the DUS definition.

2. Missing literature: symmetry, compression, hyperbolic embeddings

The authors overlook a substantial body of related work, including directed acyclic graphs (DAGs), network symmetry, compression symmetry, and hyperbolic embeddings. As examples, I would point to Krioukov et al., “Hyperbolic geometry of complex networks”, and Sánchez-García, “Exploiting symmetry in network analysis”.

Response. Accepted, and both suggested references are now cited, along with related work the reviewer’s comment led me to. The new related-work paragraph in the Introduction places this paper among approaches to meso-scale structure that do not rely on statistical inference: exact symmetry (MacArthur, Sánchez-García and Anderson; Sánchez-García), graph fibrations (Boldi and Vigna; Morone et al., already cited in the original), latent hyperbolic geometry (Krioukov et al.), and explicit hierarchy measures for directed networks (Mones et al.; Corominas-Murtra et al.).

The connection to the symmetry literature is more than bibliographic and I have made it explicit in the DPS section. Two activities in the same DPS class have predecessor sets occupying the same DUS levels, which is a local symmetry of the sort that fibrations formalise; the DPS can be read as a coarse, level-indexed relaxation of a fibration, which is what makes it computable by a single pass and labellable by one integer. Sánchez-García’s framing of symmetry as a route to network reduction is exactly the use made of it here.

Minor comments

The construction of the DUS and DFS networks needs clearer explanation; there is currently some confusion between the network itself and the associated metric/category.

Response. Accepted; this was a real ambiguity in the original text, which used “DUS” for the index, for a single level, and for the partition. The paper now names four distinct objects and uses them consistently: the *DUS index* u_i , an integer attached to an activity; a *DUS level*, the set of activities sharing one index value; the *Dependency Uprise Structure*, the partition into levels; and the *DUS network*, whose nodes are levels. The same four-way distinction is drawn for the DPS. Phrases such as “the number of DUSs” have been replaced throughout by “the number of DUS levels” or “the number of DPS classes”.

“Equation (1)” is referred to elsewhere in the text simply as “(1),” which is ambiguous on first read.

Response. Fixed. All equation references now read “Eq. (1)” or “Eq. (2)”.

Figure 1 is very difficult to read due to the complexity (number of nodes and edges) of the network, which obscures a clear understanding of the method’s benefits.

Response. Accepted, and the figure has been redrawn (it is now Figure 2). The original used a semicircular arc layout in which the level sizes, the spine and the skip dependencies were superimposed in one crowded picture. The new version separates them: the DUS spine runs along a linear horizontal axis, bars below give the number of activities per level coloured by progress, and arcs above show dependencies between non-adjacent levels, drawn as semicircles

whose radius is the span so that long-range links are visually distinct from short ones. Only the strongest 5% of skip dependencies are drawn, with the total stated on the figure so that nothing is hidden silently. Panel titles now give the activity, dependency and level counts. Both schedules shown are included in the public code repository, so the figure can be regenerated by any reader.

Reviewer 3

Overall I like the paper, but I have a few comments.

Response. I thank the reviewer for the encouragement and for a set of comments that improved the exposition considerably.

1. An example of a WBS

I am not in project management, so in the introduction it would be useful to see an example (even a simplified one) of a WBS, to get a better sense of the objects of study.

Response. Done. Figure 1A is a small WBS for a house, decomposed into substructure, superstructure and services and then into seven schedulable activities, and the Introduction now walks through it. The text notes that real construction WBSs have exactly this shape but reach tens of thousands of leaves. Figure 1B then shows the dependency network over those same seven activities, so the reader can see both objects and the fact that they do not coincide.

2. A diagram for the DUS, and the relation to breadth-first search

The definition of the DUS is clear enough, but again, it seems like a diagram would be helpful here. Perhaps move Figure 4 to this section. How does the DUS index differ from a breadth first search starting from the root ‘project completed’ node?

Response. The diagram has been moved forward as requested and now appears as Figure 1, ahead of both definitions; it is referenced from the DUS section.

The breadth-first search question was a good one and the answer is now a paragraph in the DUS section. The two differ because breadth-first search computes *shortest* path lengths while Eq. (1) computes *longest* ones. They disagree whenever an activity has predecessors at different depths. The paper gives the concrete case from Figure 1B: a breadth-first search from the start would place Utilities at depth 1, because Excavate feeds it directly, whereas Eq. (1) places it at $u = 2$ because it must also wait for Foundations. Only the longest-path rule guarantees that every one of an activity’s predecessors lies strictly below it, which is the property the whole construction depends on — it is what makes the levels ordered, what makes activities within a level pairwise incomparable, and what guarantees the DUS spine exists. A backwards breadth-first search from the completion node is also addressed: it would group activities by remaining depth to the end rather than by accumulated depth from the start, which is a different and in general incompatible grouping.

3. The name “uprise”

On naming, what does ‘uprise’ mean in this context? The usual use of the word is as a verb ‘uprising’ meaning a rebellion which I don’t think is the intention! Why not just ‘dependency structure’?

Response. The reviewer is right that the word is unusual, and the intended sense was never stated. It is now: the index counts how far an activity has risen above the start of the project, in dependency steps rather than in calendar time, and a sentence in the DUS section says so.

I have kept the name, for two reasons I hope the reviewer will find reasonable. “Dependency structure” on its own would be ambiguous, since it is a natural description of the dependency network itself, and it also collides with Design Structure Matrix terminology, which Reviewer 1 has asked me to engage with directly. And the name and its acronym are already in use in the public code repository. If the editor or the reviewer feels the term remains an obstacle, I am willing to rename it — “Dependency Level Structure” would be the natural alternative — but I did not want to change it unilaterally given the acronym is now load-bearing across the paper and the software.

4. Redundant panels in Figure 1

Figure 1 is nice, but why include A and B when the same information is in C and D?

Response. Accepted; panels A and B have been removed. The figure (now Figure 2) shows only the two annotated profiles, which as the reviewer observed contain everything the bare networks did. The freed space has been used to draw the remaining panels larger and to add the readability improvements requested by Reviewer 2.

5. The flat trend in Figure 3

I realise the point of Figure 3 is to show real DUSs fall below the line, but the data appears mostly constant indicating the same average distance to the end regardless of project size? Perhaps this is obvious in project management but seems worth commenting on.

Response. This was the single most useful observation in the reviews, and the reviewer is right. The data are nearly constant, this is not obvious in project management, and it is a stronger and more interesting statement than the one the original manuscript made. It is now the main claim of that section rather than an unremarked feature of the plot.

Quantitatively: median distance 2.5, interquartile range [1.9, 3.6], with binned medians of 3.4, 2.3 and 2.7 for projects with fewer than 50, 50–150 and more than 150 DUS levels, while m itself ranges from 14 to 571. The rank correlation with m is $\rho = 0.10$, nominally significant at $n = 1198$ but negligible. Figure 4 is now plotted on logarithmic axes with binned medians overlaid, which makes the flatness against the rising chain expectation obvious; on the original linear axes it was not. The text draws the practical consequence: a schedule with 500 DUS levels is not 500 steps deep in any operational sense, because long-range dependencies carry information to the end in two or three steps. This also connects to Reviewer 1’s point 3, where the strong small-world claim has been replaced by exactly this statement.

6. The DFS name and DPS visualisation

Regarding the name, I strongly suggest you change it since DFS=Depth First Search and this will probably cause a lot of confusion. Does the DFS admit any nice visualisations like Figure 1?

Response. Accepted without reservation — the clash is real and would have caused confusion, particularly since the paper discusses graph traversal explicitly in answering the reviewer’s point 2. The construction has been renamed throughout to **Dependency Predecessor Structure**

(DPS). The new name is also more descriptive: the grouping is defined precisely by which DUS levels an activity’s predecessors occupy. The connection to graph fibrations is retained and made more explicit in the text, but it no longer drives the acronym.

On visualisation, the DPS is now shown in Figure 1B, where DPS class is encoded by node colour, and in Figure 1C, where the DPS network is drawn alongside the DUS network so that the two coarse-grainings can be compared directly on the same example. I did not add a DPS analogue of the real-schedule profile in Figure 2. The reason is that a DPS network of a real project has of order 10^3 nodes with no canonical linear ordering — the DPS index is a categorical label, not a coordinate — so the arc-diagram layout that makes the DUS profile legible has nothing to lay out against. Finding a good visual representation for the DPS network is noted as future work.

7. Insight for project management

I assume the purpose of these constructions is to provide some insight into project management schedules. The author should discuss this.

Response. Accepted; the Discussion now has a paragraph devoted to this, giving three concrete uses. Reporting and tracking: the DUS profile compresses a schedule of tens of thousands of activities into a picture with of order a hundred nodes on which progress, cost or risk can be overlaid, and in which a stalled level or an unusual concentration of work is immediately visible. Diagnosis: the long arcs in Figure 2 are dependencies that skip many levels and therefore propagate delay furthest, which makes them natural candidates for scrutiny — such a link is usually either a real structural constraint or an artefact of how the schedule was built. Input to risk models: a coarse-grained network with a hundred nodes is tractable for Monte Carlo simulation where one with a hundred thousand is not. The paragraph also notes why determinism matters in this setting: two planners applying the construction to the same schedule get the same answer, which is not true of clustering methods that depend on an objective function or a random seed. A further paragraph now states three limitations explicitly, including that the DUS is one antichain decomposition among many and is not claimed to be optimal by any criterion.